

B6: Condensed-Matter Physics — 2015 Model Solutions

Trinity Term 2015 (Paper A13485W1)

Question 1 — Fermi/Debye temperatures and electronic vs. lattice heat capacity

Problem. Monovalent metal: n atoms (and electrons) per unit volume; sound speed c . Derive T_F and θ_D , sketch C_e , derive C_{ph} for $T \ll \theta_D$, and apply to Cu (fcc, $a = 0.36$ nm, $c = 2700$ m s⁻¹, $A = \pi^2/2$).

(a) Fermi and Debye temperatures

Free-electron sphere: each $\mathbf{k}$ -state occupies $(2\pi)^3/V$ in $\mathbf{k}$ -space; spin degeneracy 2:

$$n = \frac{2}{(2\pi)^3} \frac{4\pi k_F^3}{3}.$$

Solve for k_F :

$$k_F = (3\pi^2 n)^{1/3}.$$

Fermi energy:

$$E_F = \frac{\hbar^2 k_F^2}{2m_e} = \frac{\hbar^2}{2m_e} (3\pi^2 n)^{2/3}.$$

Definition $k_B T_F = E_F$:

$$T_F = \frac{\hbar^2}{2m_e k_B} (3\pi^2 n)^{2/3}.$$

Debye sphere fills the Brillouin zone with $3N$ phonon modes; 3 polarisations, density $V/(2\pi)^3$:

$$3n = \frac{3}{(2\pi)^3} \frac{4\pi k_D^3}{3}.$$

Solve for k_D :

$$k_D = (6\pi^2 n)^{1/3}.$$

Linear dispersion $\omega_D = c k_D$:

$$\omega_D = c (6\pi^2 n)^{1/3}.$$

Definition $k_B \theta_D = \hbar \omega_D$:

$$\theta_D = \frac{\hbar c}{k_B} (6\pi^2 n)^{1/3}.$$

(b) Electronic heat capacity — simple argument

At $T \ll T_F$, only electrons within $\sim k_B T$ of E_F are thermally active. Number of such electrons per unit volume:

$$\Delta n \sim n \frac{k_B T}{E_F} = n \frac{T}{T_F}.$$

Each gains thermal energy $\sim k_B T$:

$$u \sim \Delta n \cdot k_B T \sim n k_B T \frac{T}{T_F}.$$

Differentiate w.r.t. T :

$$C_e = \frac{\partial u}{\partial T} = A n k_B \left(\frac{T}{T_F} \right),$$

with A an $O(1)$ constant ($A = \pi^2/2$ from the exact Sommerfeld expansion).

(c) Debye assumptions and low- T phonon heat capacity

Assumptions:

- Linear, isotropic dispersion $\omega = c k$ (one effective sound speed).
- Single Debye cutoff k_D chosen so that the total number of modes equals $3N$.
- Phonons are quantised harmonic oscillators (Bose statistics).

Mode density (3 polarisations):

$$g(\omega) d\omega = \frac{3}{(2\pi)^3} \cdot 4\pi k^2 dk, \quad \omega = c k.$$

Eliminate k :

$$g(\omega) = \frac{3\omega^2}{2\pi^2 c^3}, \quad 0 \leq \omega \leq \omega_D.$$

Internal energy per unit volume:

$$u(T) = \int_0^{\omega_D} \hbar \omega n_B(\omega) g(\omega) d\omega, \quad n_B = \frac{1}{e^{\hbar \omega / k_B T} - 1}.$$

Substitute $g(\omega)$:

$$u(T) = \frac{3\hbar}{2\pi^2 c^3} \int_0^{\omega_D} \frac{\omega^3}{e^{\hbar \omega / k_B T} - 1} d\omega.$$

Change variable $x = \hbar \omega / k_B T$:

$$u(T) = \frac{3\hbar}{2\pi^2 c^3} \left(\frac{k_B T}{\hbar} \right)^4 \int_0^{\hbar \omega_D / k_B T} \frac{x^3}{e^x - 1} dx.$$

Low- T limit $T \ll \theta_D$: extend upper limit to ∞ , use $\int_0^\infty x^3 / (e^x - 1) dx = \pi^4/15$:

$$u(T) = \frac{3\hbar}{2\pi^2 c^3} \frac{(k_B T)^4}{\hbar^4} \frac{\pi^4}{15}.$$

Use $\hbar \omega_D = k_B \theta_D$ and $\omega_D = c k_D$, so $c^3 = (k_B \theta_D / \hbar)^3 / k_D^3$ and $k_D^3 = 6\pi^2 n$:

$$\frac{1}{c^3} = \frac{\hbar^3 6\pi^2 n}{(k_B \theta_D)^3}.$$

Substitute:

$$u(T) = \frac{3\hbar}{2\pi^2} \cdot \frac{\hbar^3 6\pi^2 n}{(k_B \theta_D)^3} \cdot \frac{(k_B T)^4}{\hbar^4} \cdot \frac{\pi^4}{15}.$$

Simplify:

$$u(T) = \frac{3\pi^4}{5} n k_B T \left(\frac{T}{\theta_D} \right)^3.$$

Differentiate:

$$C_{\text{ph}} = \frac{\partial u}{\partial T} = \frac{12\pi^4}{5} n k_B \left(\frac{T}{\theta_D} \right)^3 \quad (T \ll \theta_D).$$

(d) Numerical estimates for Cu

Fcc, monovalent: 4 atoms (and electrons) per conventional cell of side a :

$$n = \frac{4}{a^3} = \frac{4}{(0.36 \times 10^{-9})^3} \text{ m}^{-3}.$$

Numerator and denominator:

$$a^3 = 4.666 \times 10^{-29} \text{ m}^3.$$

Divide:

$$n = 8.57 \times 10^{28} \text{ m}^{-3}.$$

Fermi temperature: $T_F = \hbar^2 (3\pi^2 n)^{2/3} / (2m_e k_B)$.

$$3\pi^2 n = 3 \times 9.870 \times 8.57 \times 10^{28} = 2.538 \times 10^{30} \text{ m}^{-3}.$$

$$(3\pi^2 n)^{2/3} = (2.538 \times 10^{30})^{2/3} = 1.861 \times 10^{20} \text{ m}^{-2}.$$

$$\hbar^2 / (2m_e k_B) = \frac{(1.055 \times 10^{-34})^2}{2 \times 9.11 \times 10^{-31} \times 1.381 \times 10^{-23}} = 4.42 \times 10^{-16} \text{ m}^2 \text{ K}.$$

$$T_F = 4.42 \times 10^{-16} \times 1.861 \times 10^{20} \text{ K} = 8.2 \times 10^4 \text{ K}.$$

Debye temperature: $\theta_D = \hbar c (6\pi^2 n)^{1/3} / k_B$.

$$6\pi^2 n = 5.076 \times 10^{30} \text{ m}^{-3}.$$

$$(6\pi^2 n)^{1/3} = 1.719 \times 10^{10} \text{ m}^{-1}.$$

$$\hbar c / k_B = \frac{1.055 \times 10^{-34} \times 2700}{1.381 \times 10^{-23}} = 2.063 \times 10^{-8} \text{ m K}.$$

$$\theta_D = 2.063 \times 10^{-8} \times 1.719 \times 10^{10} \text{ K} = 354 \text{ K}.$$

Crossover $C_e = C_{\text{ph}}$. Equate the two with $A = \pi^2/2$:

$$\frac{\pi^2}{2} n k_B \frac{T}{T_F} = \frac{12\pi^4}{5} n k_B \left(\frac{T}{\theta_D} \right)^3.$$

Cancel $n k_B$:

$$\frac{\pi^2}{2} \frac{T}{T_F} = \frac{12\pi^4}{5} \frac{T^3}{\theta_D^3}.$$

Solve for T^2 :

$$T^2 = \frac{5}{24\pi^2} \frac{\theta_D^3}{T_F}.$$

Substitute numbers:

$$T^2 = \frac{5}{24 \times 9.870} \times \frac{(354)^3}{8.23 \times 10^4} \text{ K}^2.$$

Numerator $\theta_D^3 = 4.44 \times 10^7 \text{ K}^3$, divide by T_F :

$$\theta_D^3/T_F = 5.39 \times 10^2 \text{ K}^2.$$

Multiply by $5/(24\pi^2) = 0.02110$:

$$T^2 = 11.4 \text{ K}^2.$$

Square root:

$$T_{\times} \approx 3.4 \text{ K}.$$

C_e dominates for $T \lesssim T_{\times} \approx 3.4 \text{ K}$.

Ratio at 300 K.

$$\frac{C_e}{C_{\text{ph}}} = \frac{\pi^2/2}{12\pi^4/5} \frac{T/T_F}{(T/\theta_D)^3} = \frac{5}{24\pi^2} \frac{\theta_D^3}{T_F T^2}.$$

Substitute $T = 300 \text{ K}$:

$$\frac{C_e}{C_{\text{ph}}} = 0.02110 \times \frac{4.44 \times 10^7}{8.23 \times 10^4 \times (300)^2}.$$

Numerator $0.02110 \times 4.44 \times 10^7 = 9.37 \times 10^5$; denominator $8.23 \times 10^4 \times 9 \times 10^4 = 7.41 \times 10^9$:

$$\left. \frac{C_e}{C_{\text{ph}}} \right|_{300 \text{ K}} \approx 1.3 \times 10^{-4}.$$

Question 2 — Structure factor and powder XRD of Al, KCl, Si

Problem. Define structure factor; relate to Bragg intensity. For Al, KCl, Si (all fcc) identify the strong rings with $h^2 + k^2 + l^2 \leq 16$ and discuss whether neutrons would change the count.

(a) Structure factor

The structure factor is the Fourier amplitude of the unit-cell scattering density evaluated at a reciprocal lattice vector $\mathbf{G}_{hkl}$. For a unit cell with N atoms at fractional coordinates (x_j, y_j, z_j) and atomic form factors f_j :

$$S_{(hkl)} = \sum_{j=1}^N f_j e^{2\pi i(hx_j + ky_j + lz_j)}.$$

The Bragg peak intensity satisfies

$$I_{(hkl)} \propto |S_{(hkl)}|^2,$$

multiplied by Lorentz, polarisation, and multiplicity factors. A reflection is absent (systematic absence) when $S_{(hkl)} = 0$, which is how non-primitive lattices and specific bases are identified. X-rays scatter off electrons ($f_j \propto Z_j$ at low $|\mathbf{q}|$); neutrons scatter off nuclei (and unpaired spins) with b_j varying irregularly with isotope. Different f_j vs. b_j patterns can change which weak reflections are visible.

(b) Allowed fcc reflections with $h^2 + k^2 + l^2 \leq 16$

Lattice contribution: fcc lattice with basis at $(0, 0, 0)$, $(\frac{1}{2}, \frac{1}{2}, 0)$, $(\frac{1}{2}, 0, \frac{1}{2})$, $(0, \frac{1}{2}, \frac{1}{2})$ gives

$$S_{\text{fcc}} = f \left[1 + e^{i\pi(h+k)} + e^{i\pi(h+l)} + e^{i\pi(k+l)} \right].$$

Selection rule:

$$S_{\text{fcc}} \neq 0 \iff h, k, l \text{ all even or all odd.}$$

Enumerate (hkl) with $h^2 + k^2 + l^2 \leq 16$ satisfying the rule:

(hkl)	$h^2 + k^2 + l^2$
(111)	3
(200)	4
(220)	8
(311)	11
(222)	12
(400)	16

Six rings.

Al (single atom, f_{Al}). All six reflections survive:

Al rings: (111), (200), (220), (311), (222), (400).

KCl: K at $(0, 0, 0)$, Cl at $(\frac{1}{2}, 0, 0)$. Combining basis with fcc:

$$S_{(hkl)} = \left[f_{\text{K}} + f_{\text{Cl}} e^{i\pi h} \right] \left[1 + e^{i\pi(h+k)} + e^{i\pi(h+l)} + e^{i\pi(k+l)} \right].$$

The basis bracket gives

$$f_{\text{K}} + f_{\text{Cl}}(-1)^h = \begin{cases} f_{\text{K}} + f_{\text{Cl}}, & h \text{ even} \\ f_{\text{K}} - f_{\text{Cl}}, & h \text{ odd.} \end{cases}$$

With $Z(\text{K})=19$, $Z(\text{Cl})=17$ the K and Cl ions are both isoelectronic with Ar ($Z = 18$): $f_{\text{K}^+} \approx f_{\text{Cl}^-}$ to a good approximation. Hence for h odd, $S \approx 0$. The (111) and (311) rings (which have all-odd indices) are extinguished:

KCl strong rings: (200), (220), (222), (400).

Si: basis at $(0,0,0)$ and $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4})$ (diamond). Basis bracket:

$$1 + e^{i\pi(h+k+l)/2}.$$

Modulus squared:

$$|1 + e^{i\pi(h+k+l)/2}|^2 = \begin{cases} 4, & h+k+l \equiv 0 \pmod{4} \\ 2, & h+k+l \text{ odd} \\ 0, & h+k+l \equiv 2 \pmod{4}. \end{cases}$$

Apply to the six fcc-allowed reflections:

(hkl)	$h+k+l$	basis factor
(111)	3	2 (weak/medium)
(200)	2	0 extinct
(220)	4	4 strong
(311)	5	2 (weak/medium)
(222)	6	0 extinct
(400)	4	4 strong

The four observed (“strong”) rings are those with non-zero basis factor:

Si rings: (111), (220), (311), (400).

(c) Same with neutrons?

Al: single atom basis, no extinctions from the basis $\Rightarrow$ same six rings.

KCl: the X-ray near-extinction of h -odd rings relied on $f_{K^+} \approx f_{Cl^-}$ (both Ar-like, $Z = 18$). Neutron coherent scattering lengths are nuclear: $b_K \approx 3.7 \text{ fm}$, $b_{Cl} \approx 9.6 \text{ fm}$, so $b_K \neq b_{Cl}$:

$$S_{(hkl)}^{h \text{ odd}} \propto b_K - b_{Cl} \neq 0.$$

Hence the (111) and (311) rings reappear $\Rightarrow$ *six* rings with neutrons.

Si: the diamond extinction $h+k+l \equiv 2 \pmod{4}$ is geometric (basis at $\frac{1}{4}, \frac{1}{4}, \frac{1}{4}$) and independent of scattering length. With one element, the basis factor is the same up to a constant:

Al: 6 (unchanged); KCl: $4 \rightarrow 6$; Si: 4 (unchanged).

Question 3 — Phonon dispersion with long-range springs

Problem. Define phonon and a measurement method. For a 1D monatomic chain (period a , mass M) with Hooke’s-law couplings C_n to all neighbours, derive the equation of motion and dispersion. Then sketch and compare two specific chains.

(a) Definition and measurement

A *phonon* is the quantum of vibrational excitation of a crystal lattice: a normal-mode oscillation with wavevector $\mathbf{k}$ and angular frequency $\omega_s(\mathbf{k})$ (s a polarisation/branch label), carrying energy $\hbar\omega_s(\mathbf{k})$ and crystal momentum $\hbar\mathbf{k}$ (defined modulo a reciprocal lattice vector).

Inelastic neutron scattering. Conservation of energy and momentum:

$$\mathbf{k}_i - \mathbf{k}_f = \mathbf{G} + \mathbf{q}, \quad E_i - E_f = \pm \hbar \omega(\mathbf{q}).$$

Measure $\mathbf{k}_i$, $\mathbf{k}_f$ and the energy transfer (e.g. triple-axis spectrometer) at fixed $\mathbf{G}$, scan $\mathbf{q}$, deduce $\omega(\mathbf{q})$ along chosen high-symmetry directions.

(b) Equation of motion and dispersion

Force on atom p from atom $p + n$:

$$F_{p,n} = C_n(u_{p+n} - u_p).$$

Sum over all neighbours (force constant depends on $|n|$, $C_{-n} = C_n$):

$$M \ddot{u}_p = \sum_{n \neq 0} C_n(u_{p+n} - u_p).$$

Use $C_{-n} = C_n$ and split:

$$M \ddot{u}_p = \sum_{n > 0} C_n(u_{p+n} + u_{p-n} - 2u_p).$$

Plane-wave ansatz $u_p = U e^{i(kpa - \omega t)}$:

$$-M\omega^2 = \sum_{n > 0} C_n(e^{ikna} + e^{-ikna} - 2).$$

Combine exponentials:

$$-M\omega^2 = \sum_{n > 0} 2C_n[\cos(kna) - 1].$$

Apply $1 - \cos(x) = 2 \sin^2(x/2)$:

$$M\omega^2 = \sum_{n > 0} 4C_n \sin^2\left(\frac{nka}{2}\right).$$

Solve for ω :

$$\omega(k) = \left(\frac{4}{M}\right)^{1/2} \left[\sum_{n > 0} C_n \sin^2\left(\frac{nka}{2}\right) \right]^{1/2}.$$

(c) Two chains: nearest neighbour vs. NN+NNN

Chain 1 (NN only, C_1).

$$\omega_1(k) = 2\sqrt{C_1/M} |\sin(ka/2)|.$$

Chain 2 (C_1 , $C_2 = C_1/2$).

$$\omega_2(k) = \left(\frac{4C_1}{M}\right)^{1/2} \left[\sin^2\left(\frac{ka}{2}\right) + \frac{1}{2} \sin^2(ka) \right]^{1/2}.$$

Sound speed. Small- k expansion: $\sin(nka/2) \approx nka/2$:

$$\omega^2 \approx \frac{4}{M} \sum_{n>0} C_n \left(\frac{nka}{2} \right)^2 = \frac{(ka)^2}{M} \sum_{n>0} n^2 C_n.$$

Sound speed $v_s = \partial\omega/\partial k$ at $k \rightarrow 0$:

$$v_s = a \left(\frac{1}{M} \sum_{n>0} n^2 C_n \right)^{1/2}.$$

Chain 1: $\sum n^2 C_n = C_1$:

$$v_{s,1} = a \sqrt{C_1/M}.$$

Chain 2: $\sum n^2 C_n = C_1 + 4 \cdot (C_1/2) = 3C_1$:

$$v_{s,2} = a \sqrt{3C_1/M}.$$

Ratio:

$$\boxed{\frac{v_{s,2}}{v_{s,1}} = \sqrt{3} \approx 1.73.}$$

Maximum frequency. Chain 1: maximum at $ka = \pi$:

$$\omega_{1,\max} = 2\sqrt{C_1/M}.$$

Chain 2: $\omega_2^2(k) = (4C_1/M) [\sin^2(ka/2) + \frac{1}{2} \sin^2(ka)]$. Use $\sin(ka) = 2 \sin(ka/2) \cos(ka/2)$:

$$\omega_2^2(k) = \frac{4C_1}{M} \sin^2(ka/2) [1 + 2 \cos^2(ka/2)].$$

Let $s = \sin^2(ka/2) \in [0, 1]$, then $\cos^2(ka/2) = 1 - s$:

$$\omega_2^2 = \frac{4C_1}{M} s (3 - 2s).$$

Differentiate w.r.t. s :

$$\frac{d\omega_2^2}{ds} = \frac{4C_1}{M} (3 - 4s).$$

Stationary at $s = 3/4$:

$$\omega_{2,\max}^2 = \frac{4C_1}{M} \cdot \frac{3}{4} \cdot (3 - \frac{3}{2}) = \frac{4C_1}{M} \cdot \frac{9}{8} = \frac{9C_1}{2M}.$$

Take square root:

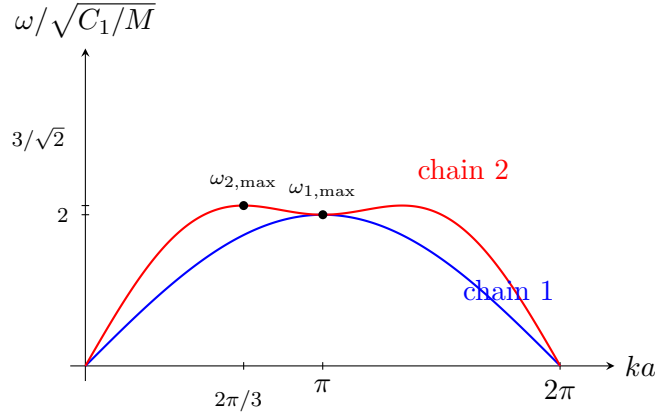
$$\omega_{2,\max} = \frac{3}{\sqrt{2}} \sqrt{C_1/M}.$$

Ratio:

$$\boxed{\frac{\omega_{2,\max}}{\omega_{1,\max}} = \frac{3/\sqrt{2}}{2} = \frac{3}{2\sqrt{2}} \approx 1.06.}$$

The maximum lies inside the BZ at $\sin^2(ka/2) = 3/4$, i.e. $ka = 2 \arcsin(\sqrt{3}/2) = 2\pi/3$.

Sketch.



Question 4 — Intrinsic and doped Si: carriers, Hall coefficient

Problem. Define intrinsic/extrinsic, derive $n(T)$ for an intrinsic semiconductor, define R_H for one carrier type, and use Hall data on Si to extract E_g , then estimate p-type doping that pins R_H over 300 K–500 K.

(a) Intrinsic vs. extrinsic

Intrinsic. Carrier concentrations determined solely by thermal excitation across the gap of the pure crystal: $n = p = n_i(T)$, E_F near mid-gap.

Extrinsic. Carriers supplied by impurity (donor/acceptor) levels: at low/intermediate T the dominant carrier type and concentration are set by the dopants, $n \neq p$, E_F pinned near the dopant level.

(b) Conduction-band electron density

Parabolic band $E(\mathbf{k}) = E_c + \hbar^2 k^2 / (2m_e^*)$. Density of states per unit volume:

$$g_c(E) = \frac{1}{2\pi^2} \left(\frac{2m_e^*}{\hbar^2} \right)^{3/2} (E - E_c)^{1/2}, \quad E \geq E_c.$$

Fermi–Dirac, non-degenerate ($E_c - E_F \gg k_B T$), Boltzmann tail:

$$f(E) \approx e^{-(E-E_F)/k_B T}.$$

Electron density:

$$n = \int_{E_c}^{\infty} g_c(E) f(E) dE.$$

Substitute $\epsilon = E - E_c$:

$$n = \frac{1}{2\pi^2} \left(\frac{2m_e^*}{\hbar^2} \right)^{3/2} e^{-(E_c-E_F)/k_B T} \int_0^{\infty} \epsilon^{1/2} e^{-\epsilon/k_B T} d\epsilon.$$

Standard integral $\int_0^{\infty} \epsilon^{1/2} e^{-\epsilon/k_B T} d\epsilon = (\sqrt{\pi}/2)(k_B T)^{3/2}$:

$$n = 2 \left(\frac{m_e^* k_B T}{2\pi \hbar^2} \right)^{3/2} e^{-(E_c-E_F)/k_B T}.$$

Define $N_c(T) = 2(m_e^* k_B T / 2\pi \hbar^2)^{3/2}$:

$$n = N_c(T) e^{-(E_c - E_F)/k_B T}.$$

Analogous derivation for holes:

$$p = N_v(T) e^{-(E_F - E_v)/k_B T}.$$

Intrinsic ($n = p \equiv n_i$): solve for E_F to eliminate it:

$$n_i^2 = np = N_c N_v e^{-E_g/k_B T}, \quad E_g = E_c - E_v.$$

Take square root, absorb prefactors into a single constant A :

$$n_i = A T^{3/2} \exp\left(\frac{-E_g}{2k_B T}\right).$$

(c) Hall coefficient and E_g from data

Steady drift in crossed $\mathbf{E}$ (along $\hat{x}$), $\mathbf{B}$ (along $\hat{z}$): transverse field cancels Lorentz force on the majority carrier. For a single carrier type with charge q (sign included) and density n_q :

$$qE_y + qv_x B_z = 0, \quad j_x = n_q q v_x.$$

Eliminate v_x :

$$E_y = -\frac{j_x B_z}{n_q q}.$$

Definition $R_H = E_y/(j_x B_z)$:

$$R_H = \frac{1}{n_q q} \left(= -\frac{1}{ne} \text{ electrons, } +\frac{1}{pe} \text{ holes} \right).$$

Why electrons dominate R_H in intrinsic Si. With both carriers present, the two-carrier Hall coefficient is

$$R_H = \frac{p\mu_h^2 - n\mu_e^2}{e(p\mu_h + n\mu_e)^2}.$$

For intrinsic Si $n = p$ and $\mu_e \approx 1350 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$, $\mu_h \approx 480 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$, so $\mu_e^2 \gg \mu_h^2$:

$$R_H \approx -\frac{1}{ne} \frac{\mu_e^2 - \mu_h^2}{(\mu_e + \mu_h)^2} \approx -\frac{1}{ne} (1 - \text{small}).$$

Hence $|R_H| \approx 1/(n_i e)$ to within $\sim 20\%$ and is well-approximated by the electron-only formula.

Extracting E_g . $|R_H| \propto 1/n_i \propto T^{-3/2} \exp(E_g/2k_B T)$:

$$\frac{|R_H(300)|}{|R_H(500)|} = \left(\frac{500}{300}\right)^{3/2} \exp\left[\frac{E_g}{2k_B} \left(\frac{1}{300} - \frac{1}{500}\right)\right] = 4 \times 10^4.$$

Compute the prefactor:

$$(500/300)^{3/2} = 2.152.$$

Take log:

$$\frac{E_g}{2k_B} \left(\frac{1}{300} - \frac{1}{500}\right) = \ln\left(\frac{4 \times 10^4}{2.152}\right) = \ln(1.86 \times 10^4) = 9.83.$$

Compute the temperature factor:

$$\frac{1}{300} - \frac{1}{500} = \frac{500 - 300}{300 \times 500} = 1.333 \times 10^{-3} \text{ K}^{-1}.$$

Solve for E_g :

$$E_g = 2k_B \frac{9.83}{1.333 \times 10^{-3}}.$$

Numerical evaluation: $9.83/1.333 \times 10^{-3} = 7.37 \times 10^3 \text{ K}$:

$$E_g = 2 \times 1.381 \times 10^{-23} \times 7.37 \times 10^3 \text{ J} = 2.04 \times 10^{-19} \text{ J}.$$

Convert to eV ($1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$):

$$E_g \approx 1.27 \text{ eV}.$$

(The accepted value for Si is 1.12 eV; the data quoted in the problem yield 1.27 eV.)

(d) Doping for a sign-flipped, T -independent R_H

Intrinsic R_H is negative (electron-like). To flip the sign and pin R_H over the whole range, dope with *acceptors* (p-type) and require $p \approx N_A \gg n_i(T)$ throughout $300 \text{ K} \leq T \leq 500 \text{ K}$. The intrinsic carrier density is largest at 500 K:

$$n_i(500 \text{ K}) = \frac{1}{|R_H(500 \text{ K})|e} = \frac{1}{(625/4 \times 10^4) \times 1.602 \times 10^{-19}}.$$

Numerator $|R_H(500 \text{ K})| = 1.5625 \times 10^{-2} \text{ m}^3 \text{ C}^{-1}$. Multiply by e :

$$|R_H|e = 1.5625 \times 10^{-2} \times 1.602 \times 10^{-19} = 2.503 \times 10^{-21} \text{ m}^3.$$

Reciprocal:

$$n_i(500 \text{ K}) = 4.0 \times 10^{20} \text{ m}^{-3}.$$

Minimum acceptor density (a comfortable factor of $\gtrsim 10$ above $n_i(500 \text{ K})$ keeps R_H flat):

$$N_A \gtrsim n_i(500 \text{ K}) \approx 4 \times 10^{20} \text{ m}^{-3} \text{ (acceptors, p-type)}.$$

Take $N_A \sim 4 \times 10^{20} \text{ m}^{-3}$ as the marginal estimate.

Electron concentration at 300 K. Mass-action $np = n_i^2$ with $p \approx N_A$, $n_i(300 \text{ K})$ from the intrinsic Hall data:

$$n_i(300 \text{ K}) = \frac{1}{|R_H(300 \text{ K})|e} = \frac{1}{625 \times 1.602 \times 10^{-19}} \text{ m}^{-3}.$$

Numerator product $625 \times 1.602 \times 10^{-19} = 1.001 \times 10^{-16}$:

$$n_i(300 \text{ K}) = 9.99 \times 10^{15} \text{ m}^{-3}.$$

Apply mass-action:

$$n(300 \text{ K}) = \frac{n_i^2(300 \text{ K})}{N_A} = \frac{(9.99 \times 10^{15})^2}{4.0 \times 10^{20}} \text{ m}^{-3}.$$

Numerator 9.98×10^{31} ; divide:

$$n(300 \text{ K}) \approx 2.5 \times 10^{11} \text{ m}^{-3}.$$